

Physical Properties of Matter

Use the cue, then fade the cue. Study the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

CUE FOCUS A **physical property** is something you can observe or measure **without** changing what the object is made of. Examples: **mass**, whether it is **attracted to a magnet**, its **relative density** (sink or float), its **solubility** (dissolves in water or not), and its ability to **conduct** heat or electricity.

Round 1 — Retrieve It

1. What is a physical property of matter?

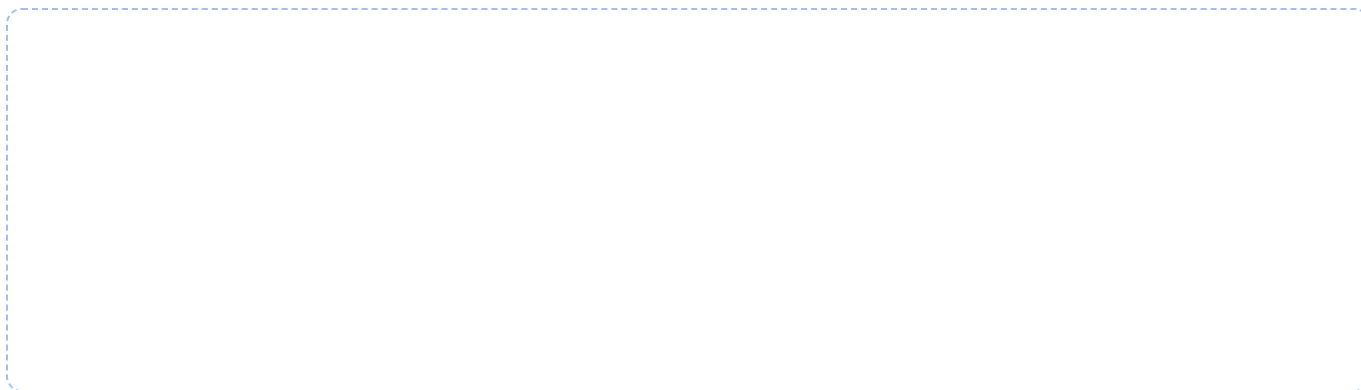
2. Name one physical property you could measure with a tool.

3. If a material dissolves in water, what property are you observing?

4. What does it mean if an object is attracted to a magnet?

Round 2 — Sketch It

Draw a cup of water with two objects in it: one that sinks and one that floats. Label which object is more dense.



Physical Properties of Matter — Apply It

Keep the cue card covered. Answer first. Then use the cue to check, revise, and explain what you changed.

Round 3 — Apply the Property

1. You have a paper clip and a plastic button. How could you use a magnet to tell them apart? What property are you testing?

2. You stir sugar into water and it disappears. Did the sugar change into something new? Explain using the word solubility.

3. You touch a metal spoon and a wooden spoon left in warm soup. Which one feels hotter, and what property explains why?

Round 4 — Reason from Evidence

A rock and a cork are the same size, but the rock sinks and the cork floats. What does this tell you about their densities? Explain your reasoning.

Physical Properties of Matter — ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain why measuring mass or testing if something is magnetic does not change what the object is made of. Give one example.

C**Connect**

Connect one physical property to a real job or tool. For example, how might someone use magnetism or density to sort materials?

E**Extend**

Design a quick test to identify a mystery material using at least two physical properties. List what you would observe or measure for each.

Confidence check. Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it.

Teacher Key — Physical Properties of Matter

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. A property you can observe or measure without changing what the object is made of.
2. Any: mass, relative density (sink or float), magnetism, solubility, ability to conduct heat or electricity.
3. Solubility (how well a material dissolves in water).
4. It is made of a magnetic material (such as iron or steel) and the magnet pulls on it.

Sketch It — look for

Two objects shown in water: one at the bottom (sinks) and one at the surface (floats). The object that sinks is labeled as more dense.

Apply It

1. Hold the magnet near both; the paper clip (metal) is attracted and the plastic button is not. The property is magnetism.
2. No new substance forms; the sugar dissolved. It is still sugar, so this shows a physical property called solubility.
3. The metal spoon feels hotter because metal conducts heat better than wood.

Reason from Evidence — look for

Same size (volume) but different results means the rock is more dense than water and the cork is less dense than water. More dense than water sinks; less dense floats.

ACE — Extend look-for

Test uses at least two physical properties (such as magnetism, density/sink-or-float, solubility, or conducting heat) and states what to observe or measure for each without changing the material.

Suggested use. Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color.